

Filling Characteristics and Dosing Accuracy of Volumetric Dosing Devices in Vertical Form-Fill-Seal Machines

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A Study has been made of the physical characteristics of the particulate products to be filled in vertical form-fill-seal machines and the configuration and dimensions of the filling and dosing devices commonly employed.

Filling from cups with different opening mechanisms and filling systems using an auger have been examined, and the dosing accuracy found in all instances to be more dependent on variations in the bulk density, general particle shape and size etc. of the product being filled than on the dimensions etc of the filling device.

Results are reported on many of the possible characteristics in an attempt to find the relevant one(s) but it is concluded that our knowledge of product characteristics is insufficient and that more research is required.

Keywords: Dosing accuracy, Vertical FFS machines, Determining dosing, Errors, Cup dosing, Auger dosing

INTRODUCTION

Intelligent machines make it possible to convert gains and losses in time made in the various basic processing operations, insofar as these gains and losses no longer have to be added to the times of other basic operations within a 360° working cycle of the main shaft. This means, of course, that studying and optimizing the basic operations will play a more central role than ever in the future development of packaging machinery construction, a realization that has determined the research work undertaken by the Fraunhofer Institute of Food Technology and Packaging in Munich for a good 10 years now. Figure 1 is a diagrammatic representation of a vertical form-fill-seal machine with its basic operations.

In this connection, of course, it is the most sensitive basic operations that are of particular

interest, i.e. those from which the greatest degree of time-saving can be expected. These are without doubt the processes of sealing the packaging and filling and dosing the product. This paper will concentrate on the latter operation, because it is these aspects above all that are hardly dealt with in the specialist literature. The state of the art, however, shows that basic findings in filling and dosing technology still offer a great potential for improvement in the constructional design of systems already in use.

The following descriptive investigations relate to commonly applied filling and dosing technologies in vertical form-fill-seal machines, the aim being to work out basic concepts which will make for an optimal configuration of processes in terms of filling speed and product accuracy, beginning

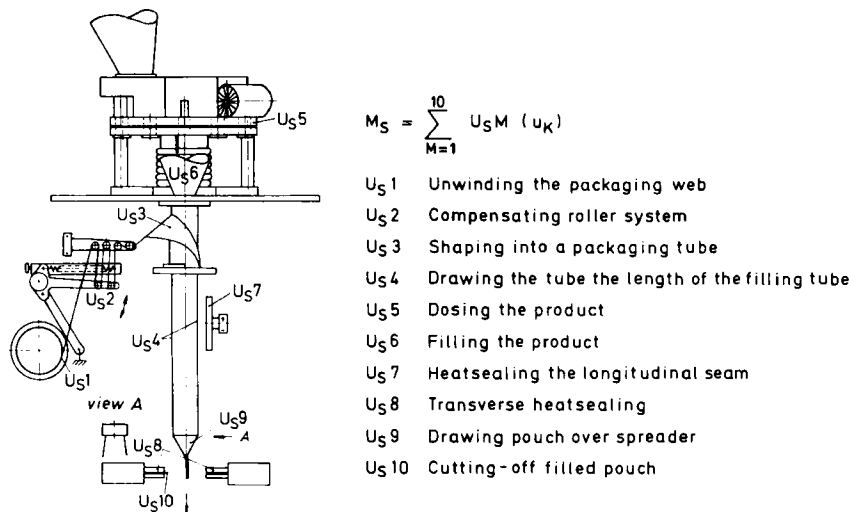


Figure 1. Vertical form-fill-seal machine providing the 10 basic operations U_{S1} to U_{S10}

with an observation of product movements and then to a study of the specific machine-related and product-related parameters that influence them. The studies were concentrated on cup dosing and auger dosing systems, i.e., volumetric filling systems.

THE BEHAVIOUR OF PRODUCTS IN CUP DOSING DEVICES

In cup dosing systems, pre-determined volumes of product are fed from a storage container into cylindrical cups which are regularly spaced around the perimeter of a dosing plate. In time with the machine cycle the cups are usually emptied through a hopper into the vertical filling tube, the product then passing directly into the packaging, which is already sealed at the bottom of the tube.

It is evident that with a rotating dosing plate the product particles that are furthest away from the axis attain greater circumferential velocities than those nearer to the axis (Figure 2). If the product falls out after being released into the cup, the particles retain their own various speeds on account of their inertia. In addition, as the particles fall downwards, they hit the wall of the hopper at different angles of impact and rebound at different speeds and angles of reflection. This results in a spiral downward movement of the product particles in the filling tube, causing the

stream to broaden, hence prolonging the actual filling time. This phenomenon can be observed very well using video techniques, which also show that although such product particle movements are similar in nature with all free-flowing products, a clear distinction must be made in quantitative terms between various types of product. This is linked with the internal friction of the product being poured into the cup, with the granular size and shape of the particles, their specific gravity, and bulk density.

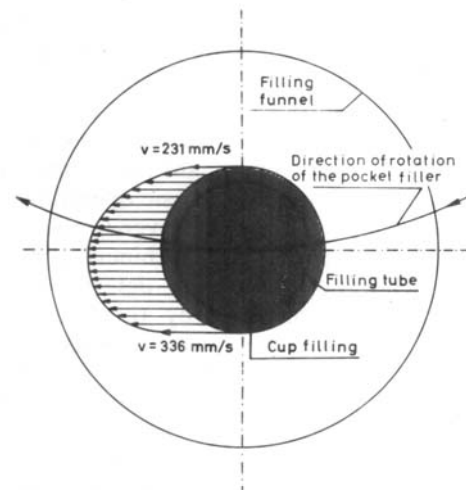


Figure 2. Translatory movement of product particles in the product dosing pockets of a pocket filler operating at 120 c/min. c = cups

A second important phenomenon has been observed by the same techniques. There is also a connection between the varying lengths of the filling-product stream and the design of the dosing flaps on the lower cup shutter. With certain types of product, if the flap opens too slowly or springs back after opening, the particles at the end of the stream will first be thrown upwards by the edge of the flap and then fall into the filling tube (Figures 3 and 4). In contrast to the phenomenon of spiral product movement, however, this problem can be solved without difficulty by suitable structural changes and therefore need not be considered further.

These observations show that the following criteria are of importance in terms of the filling times that can be achieved:

- (1) the properties of the filled product;
- (2) the height/diameter ratio of the dosing cups;
- (3) the opening mechanism;
- (4) the filling tube's inside diameter;
- (5) the geometry of the hopper;
- (6) the effectiveness of filling without a hopper;
- (7) the free drop height.

We shall now examine these factors in detail.

Equipment

The studies were made using a vertical form-fill-seal machine, the ROVEMA MVP 220, with an

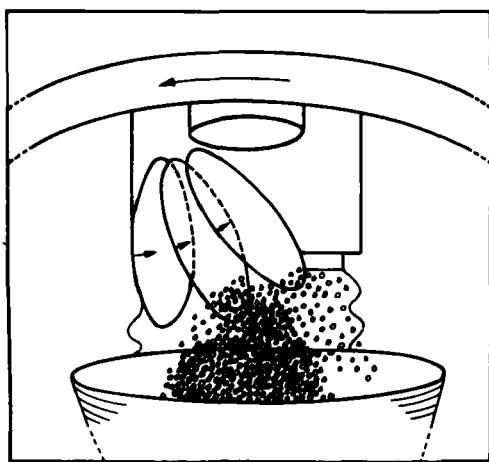


Figure 3. Swinging back of flap during filling operation (schematic)

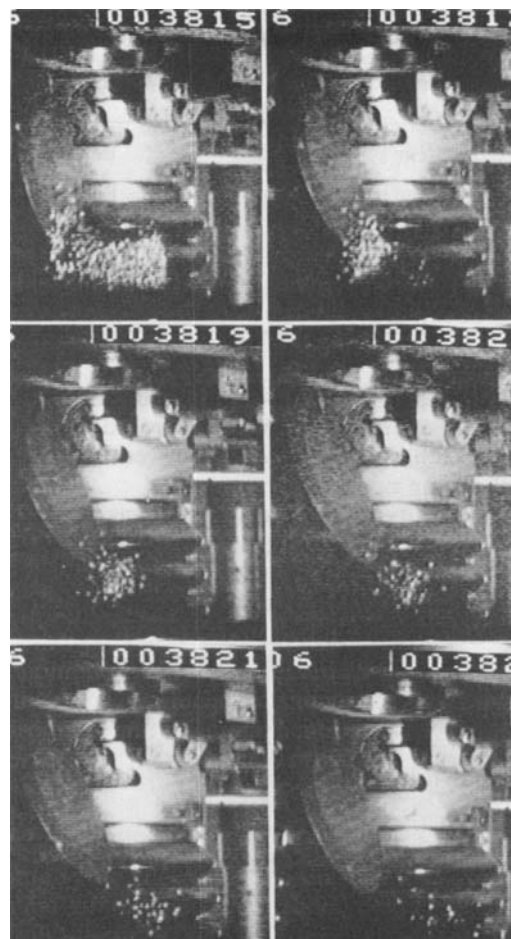


Figure 4. Swinging back of flap during filling operation with PE (polyethylene) granulate, causing particles to be hurled towards upper edge of filling funnel (standby photographs, 8.3 ms)

intermittently revolving volume-dosing device with eight telescopic dosing elements. The dosing device can be adapted for either flap dosing or slip-plate feeding. The telescopic cups were adjustable in height, and two rotating plates with different cup diameters were used so that different volumes could be filled (Table 1).

Observation of the product stream was made by means of a video camera, with which 120 pictures per second could be taken. For this purpose a transparent filling tube made of acrylic plastic was mounted in the machine. All tests were made at 60 cycles/min. The product movement is independent of the speed of the machine, however, because the rotary movement

Table 1. Data on the telescopic cups in the dosing equipment

	Wide and flat cups	Narrow and tall cups
Cup volume (cm ³)	Between 573 and 1109	Between 547 and 911
Cup height (mm)	Between 80 and 145	Between 120.5 and 185.5
Upper diameter (mm)	96.5	76
Lower diameter (mm)	102.5	84.5
Height upper diameter ratio	Between 0.84 and 1.52	Between 1.59 and 2.44

of the dosing device is independent of the number of cycles: greater machine speeds would therefore only shorten the dead times between the individual dosing operations.

The quantity of products dosed was increased in 100-ml steps. Tests were made using products with very different properties (Table 2), in terms of their bulk density, granular shape, granular size and angle of repose, but all were considered suitable to be filled in vertical form-fill-seal machines because they do not dust and they exhibit good flow properties. The codes used in Table 2 serve to identify the filling time characteristics shown later in the paper.

Filling time characteristics for cup dosing devices

The following machine parameters were examined as factors that influence the filling time:

- (a) the height/diameter ratio of the dosing cups;
- (b) the opening mechanism;
- (c) the inside diameter of the filling tube;
- (d) the geometry of the hopper;
- (e) hopper-less configuration;
- (f) the free drop height.

The height/diameter ratio of the dosing cups. The main aim of this series of tests was to determine whether, given pre-determined conditions in terms of the product, particular cup geometries were suitable for certain products. The smallest diameter used in the tests was 76 mm for the narrow cups and 96.5 mm for the wide cups (Table 1).

Tests with six different products showed (Figure 5) that it is not possible to make any universal statement, since the most favourable cup configuration depends on the type of product and the volume filled.

In the case of semolina balls (K), for instance, the wide set of cups proved to be somewhat more suitable, with a saving of up to 50 ms on filling times. The reverse was found to be true in the case of oat flakes (H), which for the larger volumes filled up to 150 ms faster. In the case of dried peas (E), the narrow cups were more favourable for volumes below 700 ml, whereas the wider set of cups was more favourable for volumes above 700 ml, producing a time-saving of up to 200 ms.

Similar ratios, if not quite so pronounced, were observed with starch pulp pellets (P), mustard seeds (F) and shelled peanuts (N) (on the right of Figure 5). It can therefore be said that, given the present state of the art, only the recording of filling time characteristics provides for a reliable optimization of filling times where the properties of the products that determine them are unknown.

Table 2. Specification and bulk density of test filling products

Product	Code	Bulk density (g/m ²)
Dried peas	E	0.276
Starch pulp pellets	P	0.646
Granulated polyethylene	G	0.581
Grated potato	A	0.342
Mustard seeds	F	0.739
Common salt	S	1.105
Fine oat flakes	H	0.340
Shelled peanuts	N	0.612
Dried leek pieces	L	0.078
Semolina balls	K	0.431

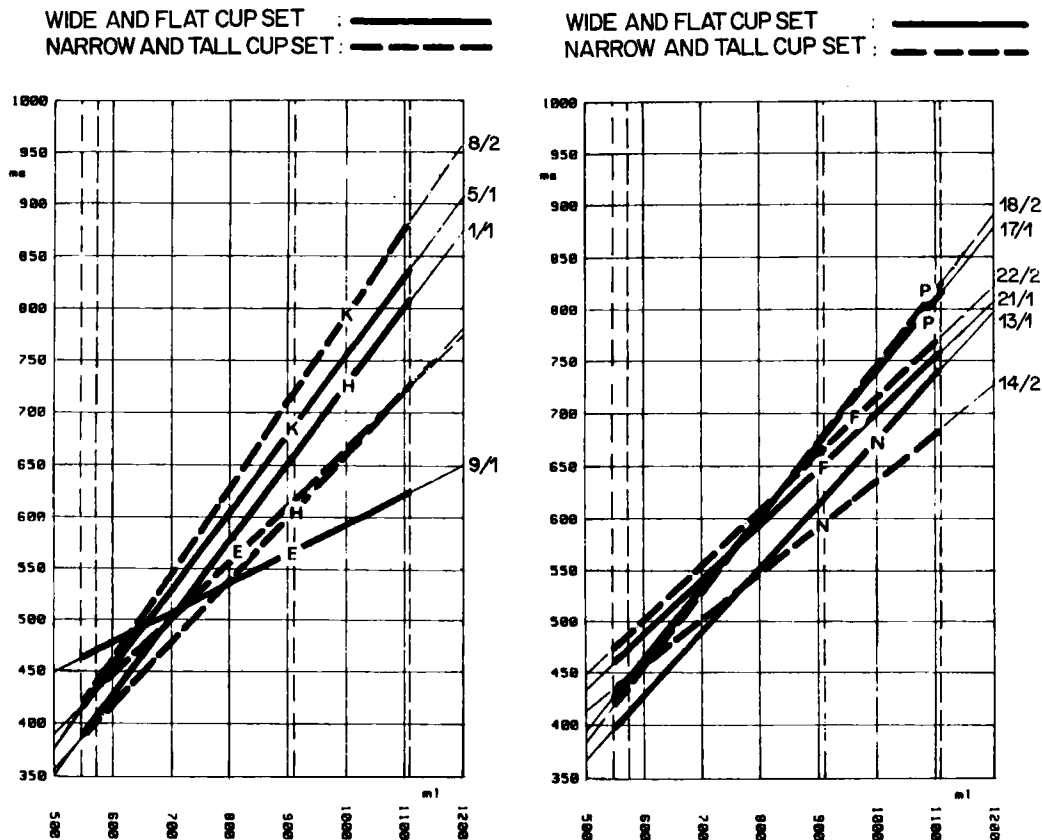


Figure 5. Filling time characteristics for flap dosing at different height/diameter ratios of dosing cups. Abscissa, filling volume; ordinate, filling time. Left: semolina dumpling (K); oat flakes (H); dried peas (E). Right: starch pulp pellets (P), mustard seeds (F), peeled peanuts (N)

Opening mechanism. Flap dosing and slip-plate feeding differ in the manner the product is emptied. As Figure 6 shows, the opening process of the flap dosing device is by means of a flap which suddenly releases the bottom end of the cup. In the case of the slip-plate feeder, the cup opening is pushed over the hopper opening as the plate revolves so that the product can be emptied gradually in accordance with the speed the plate revolves. The hoppers used in this process are much narrower, of course, than those used in the flap dosing process. In this instance the angle was 7° as opposed to 64° for the flap hopper (Figure 6).

The recorded filling time characteristics (Figure 7) show that flap dosing is advantageous particularly for products that flow easily; the filling times for common salt (S) were reduced by around 200 ms. Similar readings were recorded with

grated potato (R). This was less pronounced in the case of granulated polyethylene (G) and starch pulp pellets (P) where the volume filled again played a more significant role. It was here that the slip-plate feeder was slightly more advantageous.

In the selection of the dosing system, too, optimization has to be based on the product to be filled.

Inside diameter of the filling tube. In vertical form-fill-seal machines the width of the bag is determined by the filling tube as a component with a set format. Product-related measures with regard to the diameter of the filling tube therefore also affect the size of the bag, and hence all other consequent features relating to the distribution of the product.

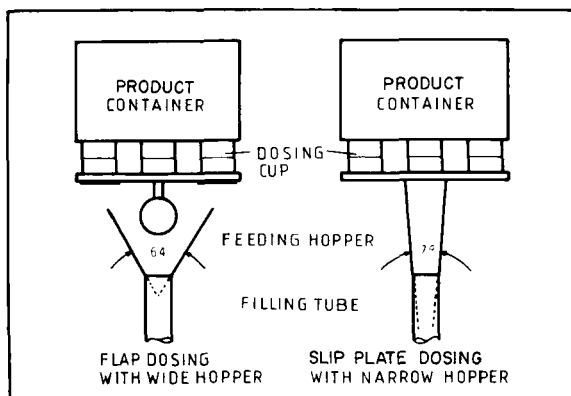


Figure 6. Arrangement of storage container, cup dosing elements, feeding hopper and filling tube for flap dosing and slip-feeder

For the tests, fixed-format components with filling tube inside diameters of 69 and 84 mm were used, corresponding to free-drop cross-sections of 37.4 and 55.4 cm², respectively.

Figure 8 shows that in this case the filling volume had a major influence on the filling time. For dried peas (E) and mustard seeds (F) the larger tube diameter proved to be more beneficial for fill volumes over approximately 900 ml, and the smaller one for volumes below 900 ml. In the case of starch pulp pellets this dividing line was reached at about 600 ml.

This means that particularly coarse products benefit from wider filling tube openings, because, on account of internal friction in the pouring process, they are exposed to greater braking resistance by internal friction.

The geometry of the hopper. Once the product has been emptied from the dosing cups, it is collected in the hopper and then fed into the filling tube opening. Because of the impact of the product particles on the hopper wall, the angle of the hopper has an important influence on the form and length of the product stream in the filling tube, as already described.¹ Hence, this angle also has a bearing on the machine speed.

In the comparative investigations, hoppers with opening angles of 64° and 7° were used, which were connected to a slip-plate feeder. Large cup sets were filled for a filling tube with an inside diameter of 84 mm.

The filling time characteristics in Figure 9 show that the narrow hopper form exhibits definite

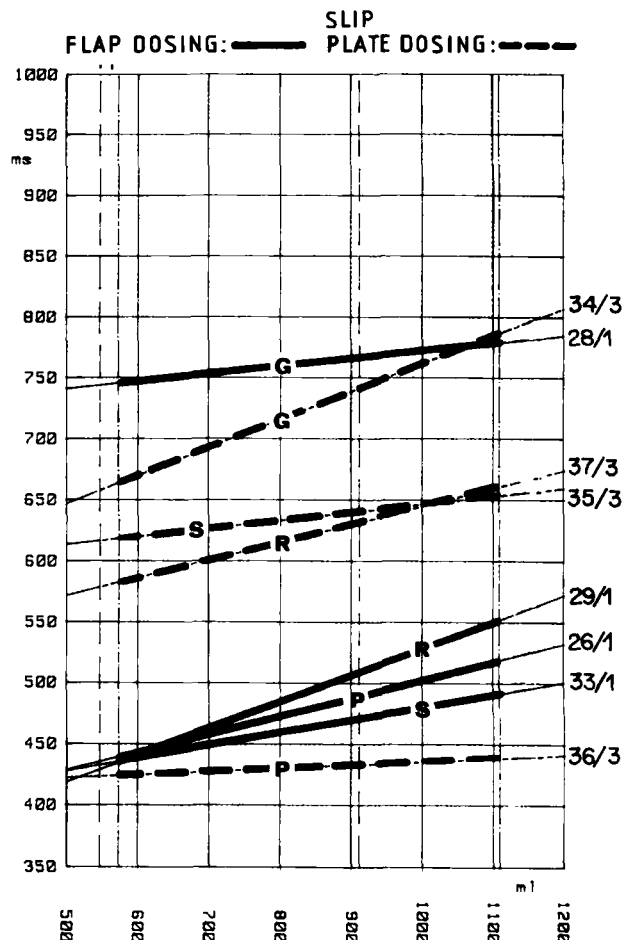


Figure 7. Filling time characteristics for flap dosing with wide feeding hopper and slip-plate feeder with narrow feeding hopper. G, granulated polyethylene; R, grated potato; P, starch pulp pellets; S, salt

advantages in each and every case. Filling time differences from 150 to almost 300 ms were recorded. This is related to the fact that in narrow hoppers the angle of impact is much smaller than in wide hoppers so that the horizontal force components that act as a brake on the product stream are considerably reduced. In practice, however, such hopper forms also mean that the overall height of the machine must be increased, unless slip-plate feeder systems are used, which could negate the gain in filling times already achieved.

Filling without a hopper. These findings prompted the idea of making tests with a hopper-less

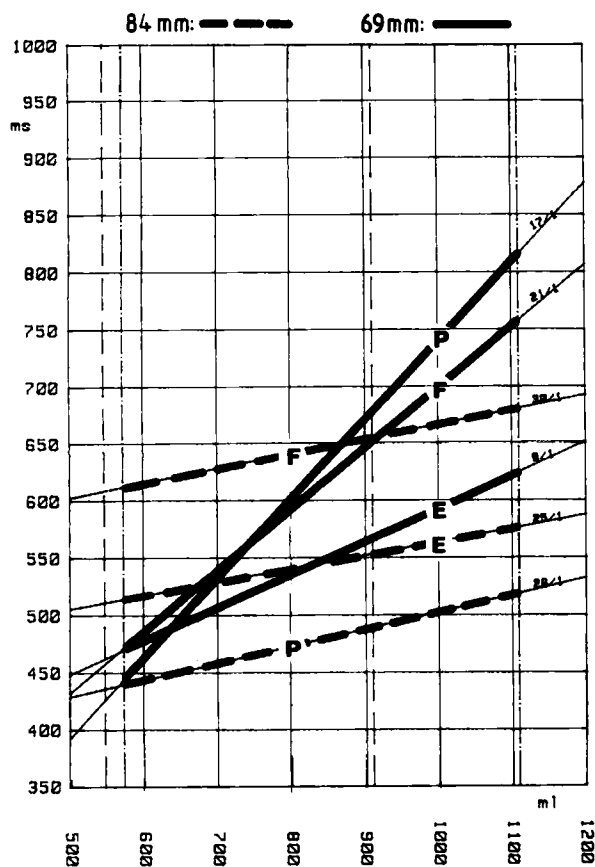


Figure 8. Filling time characteristics for flap dosing with inside diameter of filling tube of 69 mm and 84 mm. P, starch pulp pellets; E, dried peas; F, mustard seeds

configuration in order to compensate for the disadvantage of horizontal particle movements and at the same time to exert a beneficial influence on the overall height of the machine. In the test configuration the slip-plate system was positioned for this purpose directly above the filling tube with, in this case, an inside diameter of 69 mm.

The filling time characteristics in Figure 10 demonstrate quite clearly, however, that with all the test products used, both coarse and fine products, the shorter filling times were always achieved *with* a hopper. This is because the terminal velocity of the product stream is greater if there is a larger distance between the dosing device and the lower edge of the filling tube even though the overall falling time of the product stream is longer. This means that it is necessary to find the optimal ratio between the angle of the

hopper opening and the overall height of the machine.

The free-drop height. The results of the tests with a hopper-less configuration made it seem reasonable to examine the influence of the free-drop height of the product stream. To this end, the dosing system was separated from the rest of the machine and fitted with an acrylic plastic filling tube with a length of 1.5 m. The filling time characteristics for drops of 65 and 130 cm were recorded for grated potato, peanuts, common salt, granulated polyethylene and dried leek.

Figure 11 shows that the most advantageous drop height is dependent on the product. For common salt (S) and granulated polyethylene (G)

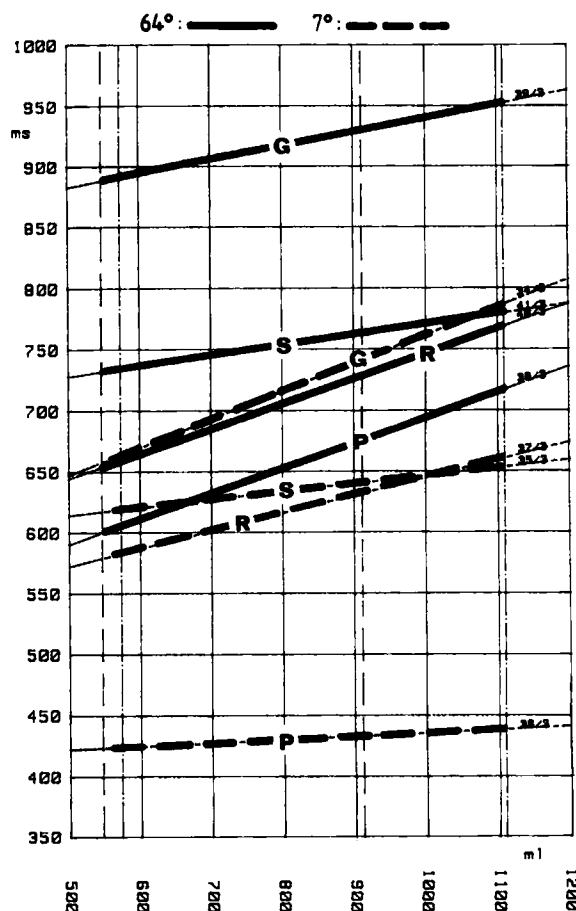


Figure 9. Filling time characteristics for slip-plate feeder at feeding hopper angles of 64° and 7°. G, granulated polyethylene; P, starch pulp pellets; S, common salt; R, grated potato

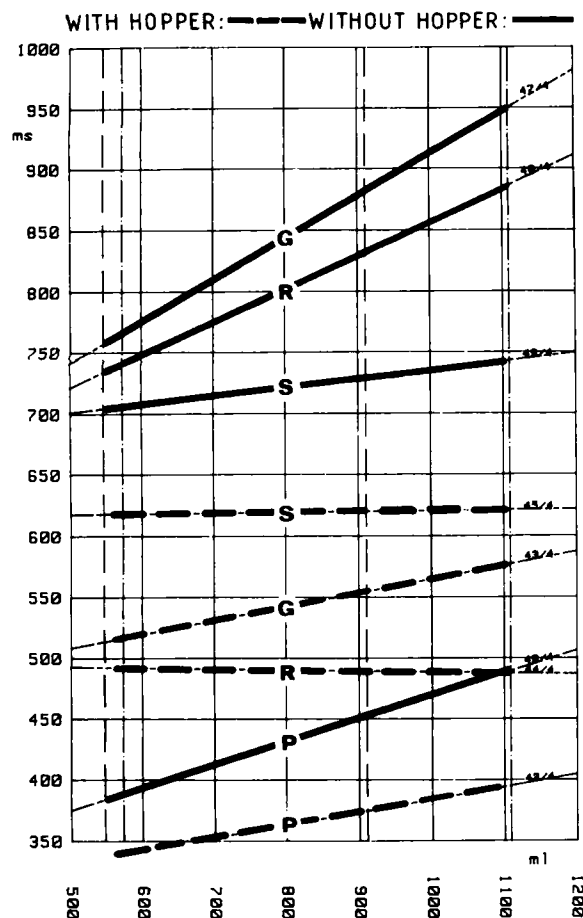


Figure 10. Filling time characteristics for slip-plate feeder, both with a narrow feeding hopper and without a feeding hopper. R, grated potato; G, granulated polyethylene; P, starch pulp pellets; S, common salt

the shorter filling times were recorded, as in the previous tests, in conjunction with the greater drop height. In the case of peanuts (N), grated potato (R) and dried leek (L), the smaller drop height proved to be more advantageous. Of decisive importance in this connection was the equilibrium between the forces exerted on the particles by the acceleration of fall and the air resistance. The lower the specific gravity of the individual particle and the larger its surface area, the sooner it is — and hence the shorter the drop — before an equilibrium with the reaction generated by air resistance is reached. Large drop heights are therefore only justified when the product particles accelerate during the entire length of the drop.

Summary. The tests showed that in terms of the filling time an optimization of equipment that apportions by volume can, at present, be achieved satisfactorily only on the basis of recording the filling time characteristics. It is above all others the manufacturers of packaging machines who can in this way gather information for the purposes of machine development. The tests revealed a lack of knowledge with respect to the product properties responsible for the differing behaviour of particles, making it impossible at present to classify products on the basis of their characteristic properties for the processes of filling and dosing.

The dosing accuracy of cup dosing devices

Besides the filling time, it is the dosing accuracy that mainly determines the success of the basic

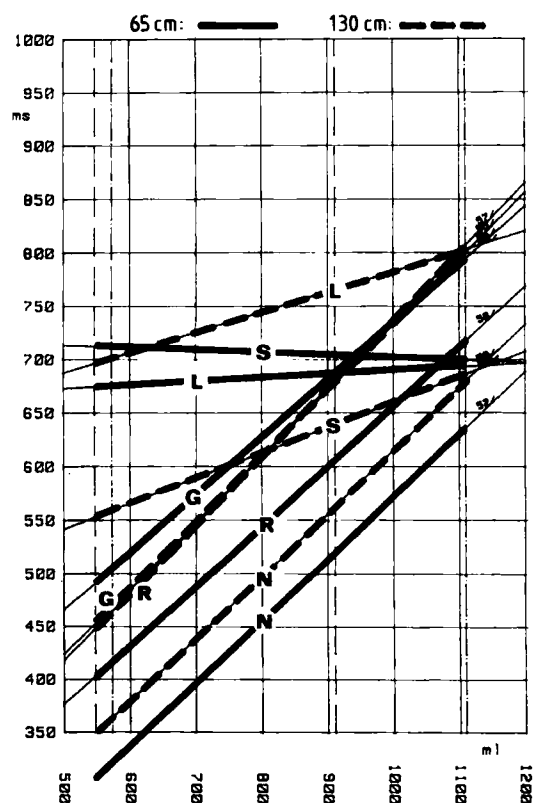


Figure 11. Filling time characteristics for slip-plate feeder for drop height of 65 and 130 cm. R, grated potato; N, shelled peanuts; S, salt; G, granulated polyethylene; L, dried leek

operation of filling and dosing. The question of dosing accuracy was examined from two aspects:

- (i) the influence exerted by the properties of the product;
- (ii) the changes in dosing accuracy caused by machine-related measures.

The influence of the nature and volume of the product. To examine the influence exerted by the nature and volume of the product, the machine was given a fixed setting in line with typical production conditions. A set of narrow cups was used with a minimum cup diameter of 76 mm; the filled product was skimmed off at the top with a plastic blade. The product was emptied from a flap dosing device into a hopper with an opening angle of 84°. The filling speed was 60/min.

Table 3 presents the recorded errors as standard deviations and variation coefficients. It demonstrates that under the given test conditions, products that are relatively homogeneous in terms of their granular distribution, like granulated polyethylene or mustard seed, produce dosing errors of less than 1%, whereas coarser products with broader granular spectrums, like peas and grated potatoes, can be filled only with

errors of 4–5%. Within the framework of this assessment, the volume of the product did not seem to have any bearing whatsoever.

It may be assumed, therefore, that as far as the product is concerned it is the regularity of the bulk density that controls the dosing accuracy, whereas the amount of product is of minor significance. At the same time, with increasing bulk density, a higher degree of dosing accuracy can be attained. In the tests we made, however, the grated potato did not follow this tendency because the repeated use of the test product led to a constant increase in the proportion of fine particles present.

The influence of machine-related parameters. In order to eliminate bulk density errors in these tests, a coarse-grain polyethylene granulate was used with an almost uniform granular size of approximately 4 mm. The bulk density (ρ_s) was 0.59 g/cm³.

In each test series, 550 and 900 cm³ of product were filled in both the set of narrow and wide cups. The product was skimmed off at the upper edge of the cup in the storage container either with a plastic blade or with a soft, rotating round

Table 3. Dosing error in dosing products of different densities with a cup dosing device

Product bulk density (ρ_s)	Filled volume (cm ³)	Filled mass (g)	Dosing error	
			Standard deviation (g)	Variation coefficient (%)
Starch pulp pellets $\rho_s = 0.63 \text{ g/cm}^3$	550	352.6	6.5	1.8
	900	565.9	6.0	1.1
Oat flakes $\rho_s = 0.38 \text{ g/cm}^3$	550	209.4	4.3	2.1
	900	340.1	7.1	2.1
Dried peas $\rho_s = 0.28 \text{ g/cm}^3$	550	153.8	6.7	4.4
	900	253.5	13.8	5.5
Grated potato $\rho_s = 0.61 \text{ g/cm}^3$	550	338.7	12.6	3.7
	900	552.3	21.0	3.8
Granulated polyethylene (fine) $\rho_s = 0.77 \text{ g/cm}^3$	550	427.3	5.5	1.3
	900	686.4	8.7	1.3
Granulated polyethylene (coarse) $\rho_s = 0.59 \text{ g/cm}^3$	550	328.7	2.4	0.7
	900	532.2	4.1	0.8
Mustard seeds $\rho_s = 0.76 \text{ g/cm}^3$	550	420.6	4.1	1.0
	900	683.8	5.9	0.9

brush. The product was extracted from the storage container either completely by means of a continuous emptying process or once 5 or 10 cups had been emptied, after which the storage container was refilled. In contrast to the complete emptying of the storage container, the partial emptying process maintains a certain level of product in the storage container between an upper and a lower mark. The difference in levels remains smaller, the fewer times the cups are filled before the storage container is refilled.

Table 4 shows the dosing errors as variation coefficients based on the set variables.

The average of the variation coefficients for the corresponding filling volumes produces dosing accuracies of 0.51% as opposed to 0.46% for filling amounts of 600 and 900 cm³, respectively.

This difference is not great, but it does confirm the tendency one would also expect in theory that larger filling volumes lead to higher degrees of accuracy.

The results shown in Table 4 do not suggest that there are any differences due to the choice of cup. The averages of the relevant variation coefficients were more or less the same. It cannot be ruled out, however, that deviations from this will be observed, depending on the product.

It is conceivable, for instance, that narrow sets of cups will allow products with a wide distribution of grain sizes to settle more readily, hence possibly exerting a negative influence on the dosing accuracy, whereas at the same time with only a small surface to be skimmed off in the same set of cups there would only be small skimming

Table 4. Impairment of dosing accuracy due to design features of cup dosing devices. The product was granulated polyethylene with a bulk density of 0.59 g/cm³

Filling volume*	Cup emptying process for the narrow cup set	Cup emptying process for the wide cup set	Dosing accuracy		
			Filling mass (g)	Standard deviation (g)	Variation coefficient (%)
600	Totally		341.7	1.9	0.55
600	10 cup contents		341.2	1.0	0.30
600	5 cup contents		343.0	0.7	0.20
900	Totally		507.9	3.1	0.60
900	10 cup contents		506.2	1.4	0.30
900	5 cup contents		505.5	1.7	0.34
600		Totally	341.2	1.9	0.56
600		10 cup contents	339.2	1.5	0.45
600		5 cup contents	340.3	1.6	0.48
900		Totally	494.1	1.8	0.37
900		10 cup contents	495.0	1.2	0.25
900		5 cup contents	494.5	1.3	0.27
600	Totally		353.6	3.6	1.01
600	10 cup contents		350.8	1.2	0.34
600	5 cup contents		350.7	1.7	0.48
900	Totally		541.5	4.5	0.82
900	10 cup contents		516.4	1.5	0.29
900	5 cup contents		539.9	2.8	0.51
600		Totally	346.3	2.1	0.62
600		10 cup contents	346.3	2.1	0.62
600		5 cup contents	336.0	2.5	0.74
600		Totally	342.5	1.5	0.44
900		10 cup contents	509.7	4.4	0.87
900		5 cup contents	511.1	2.3	0.45
900			508.0	2.1	0.41

*In the first twelve tests listed the products were skimmed off using a plastic blade, and in the latter twelve tests listed a rotating round brush was used.

errors. As far as cup configuration is concerned, the product type will doubtless play an important role with regard to dosing accuracy.

Skimming off the product at the top of the cup using a plastic blade has distinct advantages in terms of accuracy over a round brush, which, however, gives the product a higher degree of protection. The average variation coefficients of 0.39% as opposed to 0.58% in the tests were proof of this.

Finally, a great influence is exerted on the accuracy also by the level of product in the storage container. Filling the container and then emptying it completely produces an average variation coefficient of 0.68%. If the storage container is filled up again after emptying 5 or 10 cups, readings of below 0.40% are reached. It may therefore be assumed that for every product there is a specific storage level characteristic of the highest possible dosing accuracy. At the time of writing, this level must be determined empirically from case to case, because the product-related properties that are responsible for this are not known.

THE DOSING ACCURACY OF AUGER DOSING DEVICES

Auger dosing devices are needed when cup dosing can be employed only with difficulty, such as when packing dusty products like flour or powdered milk. Fragile products on the other hand are not very suited to auger filling, which likewise applies to free-flowing products because these are able to flow of their own accord out of the filler opening of the vertically-positioned auger if the machine is at a standstill.

The movement of the product in the auger is determined by the auger itself, so that there was no need to examine the product movement in this dosing device. Of much greater interest, however, was an assessment of dosing errors, namely, how they are caused by:

- (1) the product;
- (2) the machine settings.

The calculation of dosing error

Contrary to cup dosing devices, the auger dosing device lends itself to a theoretical estimate of the

dosing error if the exact dimensions of the dosing device are known.

The dosing volume V_N ejected by the dosing screw with each turn of the screw is calculated with the dimensions given in Figure 12 as

$$V_N = \left(\frac{D_1^2 - D_2^2}{4} \right) - V_s,$$

V_s being the volume of the auger itself and is calculated as follows:

$$V_s = \left(\frac{D_1}{2} - \frac{D_2}{2} \right) \cdot b \cdot \sqrt{\left[\left(\frac{D_1}{2} + \frac{D_2}{2} \right)^2 \cdot \pi^2 + s^2 \right]},$$

where b and s are the thickness and pitch of the auger screw as indicated in Figure 12. With the dimensions of the dosing device used in the tests as shown in Figure 12, a dosing volume results of 58.7 cm^3 per turn of the auger.

The ejected product amount is

$$m = V_N \cdot n \cdot \rho_s,$$

with n being the number of auger turns and ρ_s the bulk density.

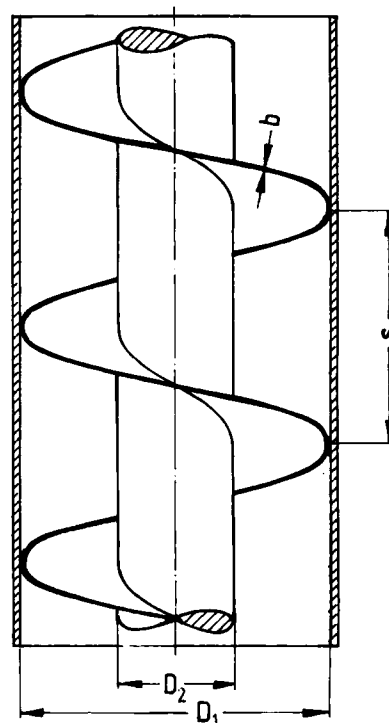


Figure 12. Dimensions of a dosing auger for the calculation of the dosing volume

According to the Gaussian law of error propagation, the following equation applies to the dosing device:

$$\frac{\Delta m}{m} = \sqrt{\left[\left(\frac{\Delta V_N}{V_N}\right)^2 + \left(\frac{\Delta n}{n}\right)^2 + \left(\frac{\Delta \rho_s}{\rho_s}\right)^2\right]}.$$

The net displacement volume, N , the number of auger turns, n , and the bulk density, ρ_s , are therefore initially the parameters responsible for dosing error.

The error in the net displacement volume is determined firstly by the geometrical dimensions of the dosing auger and is hence linked with the work tolerances of the individual elements of the machine. However, the present state of the art is such that these are so small that this term can be ignored in the error equation.

This applies similarly to the number of auger turns, since the technological means available nowadays make a very exact setting of rotations possible, which permits hardly any noticeable dosing error.

There remains then only the dosing errors caused by the bulk density, so that the above error equation can be simplified to read

$$\frac{\Delta m}{m} \approx \frac{\Delta \rho_s}{\rho_s}$$

The determination of dosing error by means of measuring techniques

The determination of the dosing error was undertaken using an auger filler, the ROVEMA DOS 377, under production conditions. The auger was driven at 500 and 1000 r.p.m. The filling volume was 289, 578 and 1156 cm³ corresponding to 5, 10 and 20 turns, respectively. At the auger outlet, mechanical outlet retarders were tested in terms of their effectiveness in maintaining dosing accuracy.

Such elements (see Figure 13) are meant to stop the product escaping from the auger when it is not turning. For this reason, the mechanical resistance between the product and the outlet retarder must be controlled in such a way that on the one hand the product is not allowed to escape and on the other hand there is no blockage of the outlet.

In the tests, grated potato ($\rho_s = 0.485 \text{ g/cm}^3$ without fine particles) and potato flakes ($\rho_s =$

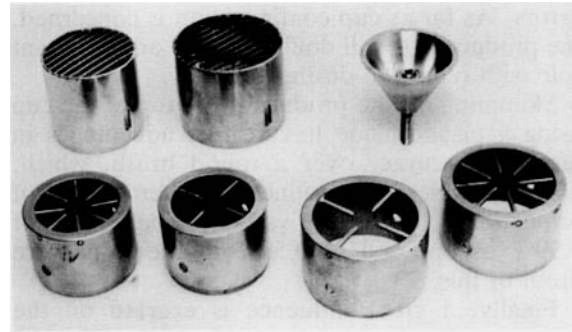


Figure 13. Outlet retarders for auger filling equipment

0.770 g/cm^3) were used. Each test was made with 60 fillings.

Table 5 shows the averages of the dosing errors recorded in more than 2000 individual fillings, again represented as the variation coefficient of the set filling weights.

Although it is evident that in spite of the large number of readings it is almost impossible to make any generalized statement, it is noticeable that the most inexact fillings were produced by grated potato. This must be seen as being directly connected with the extremely heavy fluctuations in the bulk densities of this product; subsequent measurements showed that during the tests the bulk density increased from the original 0.485 g/cm^3 to 0.603 g/cm^3 , in other words by around 20%. The escape of the fine particles was most readily controlled with a four-pin outlet retarder, although only in the lower range of turns, while the outlet became blocked when the product was transported quicker:

The dosing accuracy proved to be much greater in the case of the generally more flaky mash-potato flakes, for which the pneumatically controlled conical shutter proved to be most efficient at the outlet end, whereas the strip inset was not always able to stop the fine particles escaping.

The results produced do not permit us to draw any universal conclusions as to the influence of the filling speed, the effectivity of outlet retarders, the size of the dosing volume or the type of product. This means that there is hardly any alternative but to ascertain the optimal correspondence between product and machine setting on an empirical basis until product-specific parameters can be used.

Table 5. Dosing error in auger dosing processes

Product	Outlet retarder	Variation coefficient (%)					
		At No. Auger turns required for filling volume					
		5		10		20	
		500 (turns/min)	1000 (turns/min)	500 (turns/min)	1000 (turns/min)	500 (turns/min)	1000 (turns/min)
Grated potato	Open	3.8	*	3.2	•	2.5	•
	4-pin	2.5	†	1.9	†	3.8	†
	conical shutter	4.2	*	4.4	*	7.5	•
Potato flakes	Open	1.9	1.6	1.1	1.1	2.9	0.8
	strip inset	††	2.0	††	2.6	0.6	3.7
	conical shutter	1.1	1.2	1.0	2.1	1.2	0.8
Granulated polyethylene	Conical shutter	2.3	4.1	2.1	3.3	2.3	2.8

*Result inaccurate because of fine particles escaping.

†Blockage in the outlet.

††Product escapes from the outlet.

SUMMARY

If we now attempt to make a final evaluation of the test results, we come to the definite conclusion that a knowledge of the products' physical properties that are characteristic of the products movements is the essential prerequisite for controlling the filling and dosing behaviour in filling and dosing devices that apportion by volume.

It is with regard to the question of dosing accuracy that the information provided by the test results is distorted by bulk density fluctuations, which either must be regarded as typical of the product itself or begin to materialize only during the processes of filling and dosing. But the bulk density was not the only characteristic variable. The granular shape and size, internal friction, the angle of repose, to name but a few, are characteristic product-related parameters that in connection with the optimization of the basic operation of filling and dosing require systematic, scientific investigation.

Of course, even now it is possible, as we have seen, to assess certain optimizing procedures with

regard to the length of the filling tube, the form of the hopper, the configuration and dimensions of the cups, the question of auger speeds, the effectiveness of outlet retarders in auger filling, the methods of emptying the storage container and so on. However, each individual decision will continue to depend upon a comparatively painstaking preliminary test providing certainty as to whether the design measures were the right ones for both the filling process and dosing accuracy.

For this reason, one of the main targets of future research projects will be to close the gaps in our knowledge of the characteristic properties of individual products.

REFERENCE

1. Vogelpohl, H. and Johmann, H. J., Maschinenanalytische Untersuchungen am Beispiel einer vertikalen Schlauchbeutelmaschine (Machine-analytical investigations, using the example of a vertical form-fill-seal machine) *Verp.-Rdsch.*, **35** (9) (1984), Techn.-wiss. Beilage S. 55/61.